PGCP-AC AUGUST 26
Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:

- Books
- Authors
- Members
- Borrow_Records

2. Write the SQL command to create a table Authors with the following fields:

- author_id (Primary Key, INT)
- name (VARCHAR(100))
- country (VARCHAR(50))

Output :-

mysql> CREATE DATABASE LibraryDB;
Query OK, 1 row affected (0.00 sec)

mysql> use LibraryDB;
Database changed

mysql> use LibraryDB;
Database changed

mysql> CREATE TABLE Author(author_id INT PRIMARY KEY AUTO_INCREMENT ,
aurthor_name VARCHAR(100) NOT NULL,
aurthor_country VARCHAR(50) NOT NULL);

Query OK, 0 rows affected (0.01 sec)

```
mysql> DESC Authors;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| author_id       | int          | NO   | PRI | NULL    | auto_increment |
| aurthor_name    | varchar(100) | NO   |     | NULL    |                |
| aurthor_country | varchar(50)  | NO   |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
3 rows in set (0.00 sec)

mysql>
```

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3. Write the SQL command to create a table Books with the following fields:
- book_id (Primary Key, INT)
- title (VARCHAR(150))
- author_id (Foreign Key referencing Authors)
- published_year (YEAR)
- available_copies (INT)

```
mysql> CREATE TABLE Books(
   -> book_id INT PRIMARY KEY AUTO_INCREMENT,
   -> title VARCHAR(150),
   -> author_id INT,
   -> published_year YEAR,
   -> available_copies INT,
   -> FOREIGN KEY (author_id) REFERENCES Authors(author_id));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DESC BOOKS;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| book_id          | int          | NO   | PRI | NULL    | auto_increment |
| title            | varchar(150) | YES  |     | NULL    |                |
| author_id        | int          | YES  | MUL | NULL    |                |
| published_year   | year         | YES  |     | NULL    |                |
| available_copies | int          | YES  |     | NULL    |                |
| genre            | varchar(50)  | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

4. Write the SQL command to create a table Members with:
- member_id (Primary Key, INT)
- name (VARCHAR(100))
- email (VARCHAR(100), unique)
- phone (VARCHAR(15))

```
mysql> CREATE TABLE Members(
   -> member_id INT PRIMARY KEY AUTO_INCREMENT,
   -> name VARCHAR(100),
   -> email VARCHAR(100) unique,
```

-> phone VARCHAR(15));

Query OK, 0 rows affected (0.02 sec)

```
mysql> DESC Members;
+-----------+--------------+------+-----+---------+----------------+
| Field     | Type         | Null | Key | Default | Extra          |
+-----------+--------------+------+-----+---------+----------------+
| member_id | int          | NO   | PRI | NULL    | auto_increment |
| name      | varchar(100) | YES  |     | NULL    |                |
| email     | varchar(100) | YES  | UNI | NULL    |                |
| phone     | varchar(15)  | YES  |     | NULL    |                |
+-----------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

5. Write the SQL command to create a table Borrow_Records with:
- record_id (Primary Key, INT)
- member_id (Foreign Key referencing Members)
- book_id (Foreign Key referencing Books)
- borrow_date (DATE)
- return_date (DATE)

mysql> CREATE TABLE Borrow_Records(

-> record_id INT PRIMARY KEY AUTO_INCREMENT,

-> member_id INT,

-> book_id INT,

-> borrow_date DATE,

-> return_date DATE,

-> FOREIGN KEY (member_id) REFERENCES Members(member_id),

-> FOREIGN KEY (book_id) REFERENCES Books(book_id));

Query OK, 0 rows affected (0.03 sec)

```
mysql> DESC Borrow_Records;
+-------------+------+------+-----+---------+----------------+
| Field       | Type | Null | Key | Default | Extra          |
+-------------+------+------+-----+---------+----------------+
| record_id   | int  | NO   | PRI | NULL    | auto_increment |
| member_id   | int  | YES  | MUL | NULL    |                |
| book_id     | int  | YES  | MUL | NULL    |                |
| borrow_date | date | YES  |     | NULL    |                |
| return_date | date | YES  |     | NULL    |                |
+-------------+------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

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6. Modify the Books table to add a column genre of type VARCHAR(50).

```
mysql> ALTER TABLE Books
   -> ADD COLUMN genre VARCHAR(50);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC BOOKS;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| book_id          | int          | NO   | PRI | NULL    | auto_increment |
| title            | varchar(150) | YES  |     | NULL    |                |
| author_id        | int          | YES  | MUL | NULL    |                |
| published_year   | year         | YES  |     | NULL    |                |
| available_copies | int          | YES  |     | NULL    |                |
| genre            | varchar(50)  | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

```
mysql> DROP TABLE Borrow_records;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW TABLES FROM LibraryDB;
+---------------------+
| Tables_in_librarydb |
+---------------------+
| authors             |
| books               |
| members             |
+---------------------+
3 rows in set (0.00 sec)
```

8. Insert 3 records into the Authors table.

```
mysql> INSERT INTO Authors(author_name,author_country)
   -> values
   -> ('Rohit','INDIA'),
   -> ('Aditya','INDIA'),
```

-> ('Aniket','INDIA');
Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Authors;
+-----------+-------------+----------------+
| author_id | author_name | author_country |
+-----------+-------------+----------------+
|         1 | Rohit       | INDIA          |
|         2 | Aditya      | INDIA          |
|         3 | Aniket      | INDIA          |
+-----------+-------------+----------------+
3 rows in set (0.00 sec)
```

9. Insert 5 books into the Books table.

mysql> INSERT INTO Books

-> (title, author_id, published_year, available_copies, genre)

-> VALUES

-> ('The Great Gatsby', 1, 1925, 5, 'Novel'),

-> ('Wings of Fire', 2, 1999, 10, 'Biography'),

-> ('The Alchemist', 3, 1988, 7, 'Fiction'),

-> ('Atomic Habits', 1, 2018, 12, 'Self-help'),

-> ('The Discovery of India', 2, 1946, 4, 'History');

Query OK, 5 rows affected (0.00 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Books;
+---------+----------------------+-----------+----------------+------------------+-----------+
| book_id | title                | author_id | published_year | available_copies | genre     |
+---------+----------------------+-----------+----------------+------------------+-----------+
|       1 | The Great Gatsby     |         1 |           1925 |                5 | Novel     |
|       2 | Wings of Fire        |         2 |           1999 |               10 | Biography |
|       3 | The Alchemist        |         3 |           1988 |                7 | Fiction   |
|       4 | Atomic Habits        |         1 |           2018 |               12 | Self-help |
|       5 | The Discovery of India |         2 |           1946 |                4 | History   |
+---------+----------------------+-----------+----------------+------------------+-----------+
5 rows in set (0.00 sec)
```

10. Insert 3 members into the Members table.

mysql> INSERT INTO Members (name, email, phone)

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-> VALUES

-> ('Rahul', 'rahul@gmail.com', '9876543210'),

-> ('Priya', 'priya@gmail.com', '9876543211'),

-> ('Amit', 'amit@gmail.com', '9876543212');

Query OK, 3 rows affected (0.00 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Members;
+-----------+--------------+-----------------------+----------------+
| member_id | name         | email                 | phone          |
+-----------+--------------+-----------------------+----------------+
|         1 | Rohit Patil  | RohitPatil@gamil.com  | +91 8485956555 |
|         2 | Rohit Ghanvat| RohitGhanvat@gamil.com| +91 8489998555 |
|         9 | Rahul        | rahul@gmail.com       | 9876543210     |
|        10 | Priya        | priya@gmail.com       | 9876543211     |
|        11 | Amit         | amit@gmail.com        | 9876543212     |
+-----------+--------------+-----------------------+----------------+
5 rows in set (0.00 sec)
```

11. Insert 4 borrow records into the Borrow_Records table.

mysql> INSERT INTO Borrow_Records

-> (member_id, book_id, borrow_date, return_date)

-> VALUES

-> (1, 1, '2026-10-01', '2026-10-05'),

-> (2, 2, '2026-10-02', '2026-10-06'),

-> (9, 3, '2026-10-03', NULL),

-> (10, 4, '2026-10-04', NULL);

Query OK, 4 rows affected (0.01 sec)

Records: 4 Duplicates: 0 Warnings: 0

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```
mysql> SELECT * FROM Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|         1 |         1 |       1 | 2026-10-01  | 2026-10-05  |
|         2 |         2 |       2 | 2026-10-02  | 2026-10-06  |
|         3 |         9 |       3 | 2026-10-03  | NULL        |
|         4 |        10 |       4 | 2026-10-04  | NULL        |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
```

12. Write an SQL query to select all books where published_year is after 2015.

```
mysql> SELECT * FROM Books
    -> WHERE published_year > 2015;
+---------+--------------+-----------+----------------+-----------------+-----------+
| book_id | title        | author_id | published_year | available_copies | genre     |
+---------+--------------+-----------+----------------+-----------------+-----------+
|       4 | Atomic Habits |         1 |           2018 |              12 | Self-help |
+---------+--------------+-----------+----------------+-----------------+-----------+
1 row in set (0.00 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

1. Parent table — Authors

```
CREATE TABLE Authors (
   author_id INT PRIMARY KEY AUTO_INCREMENT,
   author_name VARCHAR(100)
);
```

2. Child table — Books

```
CREATE TABLE Books (
   book_id INT PRIMARY KEY AUTO_INCREMENT,
   title VARCHAR(150),
   author_id INT,
   FOREIGN KEY (author_id)
      REFERENCES Authors(author_id)
);
```

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14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

```
mysql> SELECT Members.name
    -> FROM Members
    -> JOIN Borrow_Records
    ->     ON Members.member_id = Borrow_Records.member_id
    -> JOIN Books
    ->     ON Borrow_Records.book_id = Books.book_id
    -> WHERE Books.title = 'Database Systems';
+-------+
| name  |
+-------+
| Rahul |
+-------+
1 row in set (0.00 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

```
mysql> SELECT * FROM Books;
+---------+-------------------------+-----------+----------------+------------------+-----------+
| book_id | title                   | author_id | published_year | available_copies | genre     |
+---------+-------------------------+-----------+----------------+------------------+-----------+
|       1 | The Great Gatsby        |         1 |           1925 |                5 | Novel     |
|       2 | Wings of Fire           |         2 |           1999 |               10 | Biography |
|       3 | The Alchemist           |         3 |           1988 |                7 | Fiction   |
|       4 | Atomic Habits           |         1 |           2018 |               12 | Self-help |
|       5 | The Discovery of India  |         2 |           1946 |                4 | History   |
|       6 | Database Systems        |         1 |           2020 |                5 | Education |
+---------+-------------------------+-----------+----------------+------------------+-----------+
6 rows in set (0.00 sec)

mysql> UPDATE Books
    -> SET available_copies = available_copies - 1
    -> WHERE book_id = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELCT * FROM Books;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
mysql> SELECT * FROM Books;
+---------+-------------------------+-----------+----------------+------------------+-----------+
| book_id | title                   | author_id | published_year | available_copies | genre     |
+---------+-------------------------+-----------+----------------+------------------+-----------+
|       1 | The Great Gatsby        |         1 |           1925 |                5 | Novel     |
|       2 | Wings of Fire           |         2 |           1999 |               10 | Biography |
|       3 | The Alchemist           |         3 |           1988 |                6 | Fiction   |
|       4 | Atomic Habits           |         1 |           2018 |               12 | Self-help |
|       5 | The Discovery of India  |         2 |           1946 |                4 | History   |
|       6 | Database Systems        |         1 |           2020 |                5 | Education |
+---------+-------------------------+-----------+----------------+------------------+-----------+
6 rows in set (0.00 sec)
```

16. Delete a record from Members where member_id = 3.

```
mysql> SELECT * FROM Members;
+-----------+---------------+-----------------------+----------------+
| member_id | name          | email                 | phone          |
+-----------+---------------+-----------------------+----------------+
|         1 | Rohit Patil   | RohitPatil@gamil.com  | +91 8485956555 |
|         2 | Rohit Ghanvat | RohitGhanvat@gamil.com| +91 8489998555 |
|         9 | Rahul         | rahul@gmail.com       | 9876543210     |
|        10 | Priya         | priya@gmail.com       | 9876543211     |
|        11 | Amit          | amit@gmail.com        | 9876543212     |
+-----------+---------------+-----------------------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members
    -> WHERE member_id = 9;
+-----------+-------+----------------+------------+
| member_id | name  | email          | phone      |
+-----------+-------+----------------+------------+
|         9 | Rahul | rahul@gmail.com | 9876543210 |
+-----------+-------+----------------+------------+
1 row in set (0.00 sec)

mysql> DELETE FROM Members WHERE member_id = 9;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (
`librarydb`.`borrow_records`, CONSTRAINT `borrow_records_ibfk_1` FOREIGN KEY (`member_id`)
 REFERENCES `members` (`member_id`))
mysql> DELETE FROM Borrow_Records
    -> WHERE member_id = 9;
Query OK, 2 rows affected (0.00 sec)

mysql> DELETE FROM Members
    -> WHERE member_id = 9;
Query OK, 1 row affected (0.00 sec)

mysql>
```

```
mysql> SELECT * FROM Members;
+-----------+---------------+-----------------------+----------------+
| member_id | name          | email                 | phone          |
+-----------+---------------+-----------------------+----------------+
|         1 | Rohit Patil   | RohitPatil@gamil.com  | +91 8485956555 |
|         2 | Rohit Ghanvat | RohitGhanvat@gamil.com| +91 8489998555 |
|        10 | Priya         | priya@gmail.com       | 9876543211     |
|        11 | Amit          | amit@gmail.com        | 9876543212     |
+-----------+---------------+-----------------------+----------------+
4 rows in set (0.00 sec)

mysql>
```

17. Update a Book name record from Book table with id = 1.

```
mysql> SELECT * FROM Books;
+---------+-----------------------+-----------+----------------+------------------+-----------+
| book_id | title                 | author_id | published_year | available_copies | genre     |
+---------+-----------------------+-----------+----------------+------------------+-----------+
|       1 | The Great Gatsby      |         1 |           1925 |                5 | Novel     |
|       2 | Wings of Fire         |         2 |           1999 |               10 | Biography |
|       3 | The Alchemist         |         3 |           1988 |                6 | Fiction   |
|       4 | Atomic Habits         |         1 |           2018 |               12 | Self-help |
|       5 | The Discovery of India |         2 |           1946 |                4 | History   |
|       6 | Database Systems      |         1 |           2020 |                5 | Education |
+---------+-----------------------+-----------+----------------+------------------+-----------+
6 rows in set (0.00 sec)

mysql> UPDATE Books SET title = 'Great Gatsby Novel'
    -> WHERE book_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Books;
+---------+-----------------------+-----------+----------------+------------------+-----------+
| book_id | title                 | author_id | published_year | available_copies | genre     |
+---------+-----------------------+-----------+----------------+------------------+-----------+
|       1 | Great Gatsby Novel    |         1 |           1925 |                5 | Novel     |
|       2 | Wings of Fire         |         2 |           1999 |               10 | Biography |
|       3 | The Alchemist         |         3 |           1988 |                6 | Fiction   |
|       4 | Atomic Habits         |         1 |           2018 |               12 | Self-help |
|       5 | The Discovery of India |         2 |           1946 |                4 | History   |
|       6 | Database Systems      |         1 |           2020 |                5 | Education |
+---------+-----------------------+-----------+----------------+------------------+-----------+
6 rows in set (0.00 sec)
```

18. Write an SQL query to list all books along with their authors' names.

```
FROM Books
JOIN Authors
ON Authors.Author_i' at line 1
mysql> SELECT Books.title, Authors.author_name
    -> FROM Books
    -> JOIN Authors
    -> ON Books.author_id = Authors.author_id;
+-----------------------+-------------+
| title                 | author_name |
+-----------------------+-------------+
| Great Gatsby Novel    | Rohit       |
| Atomic Habits         | Rohit       |
| Database Systems      | Rohit       |
| Wings of Fire         | Aditya      |
| The Discovery of India | Aditya      |
| The Alchemist         | Aniket      |
+-----------------------+-------------+
6 rows in set (0.00 sec)
```

SELECT columns

FROM first_table

JOIN second_table

ON first_table.common_id = second_table.common_id;

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

```
mysql> SELECT * FROM Books;
+---------+-------------------------+-----------+----------------+-------------------+-----------+
| book_id | title                   | author_id | published_year | available_copies | genre     |
+---------+-------------------------+-----------+----------------+-------------------+-----------+
|       1 | Great Gatsby Novel      |         1 |           1925 |                 5 | Novel     |
|       2 | Wings of Fire           |         2 |           1999 |                10 | Biography |
|       3 | The Alchemist           |         3 |           1988 |                 6 | Fiction   |
|       4 | Atomic Habits           |         1 |           2018 |                12 | Self-help |
|       5 | The Discovery of India  |         2 |           1946 |                 4 | History   |
|       6 | Database Systems        |         1 |           2020 |                 5 | Education |
+---------+-------------------------+-----------+----------------+-------------------+-----------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Books
    -> WHERE published_year < 2000;
+---------+-------------------------+-----------+----------------+-------------------+-----------+
| book_id | title                   | author_id | published_year | available_copies | genre     |
+---------+-------------------------+-----------+----------------+-------------------+-----------+
|       1 | Great Gatsby Novel      |         1 |           1925 |                 5 | Novel     |
|       2 | Wings of Fire           |         2 |           1999 |                10 | Biography |
|       3 | The Alchemist           |         3 |           1988 |                 6 | Fiction   |
|       5 | The Discovery of India  |         2 |           1946 |                 4 | History   |
+---------+-------------------------+-----------+----------------+-------------------+-----------+
4 rows in set (0.00 sec)
```

20. Write an SQL query to **find all books that are never borrowed (i.e., no records in** `Borrow_Records`**).**

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```
mysql> SELECT Books.*
    -> FROM Books
    -> LEFT JOIN Borrow_Records
    -> ON Books.book_id = Borrow_Records.book_id
    -> WHERE Borrow_Records.book_id IS NULL;
+---------+-----------------------+-----------+----------------+------------------+-----------+
| book_id | title                 | author_id | published_year | available_copies | genre     |
+---------+-----------------------+-----------+----------------+------------------+-----------+
|       3 | The Alchemist         |         3 |           1988 |                6 | Fiction   |
|       5 | The Discovery of India |         2 |           1946 |                4 | History   |
|       6 | Database Systems      |         1 |           2020 |                5 | Education |
+---------+-----------------------+-----------+----------------+------------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM Books b
    -> WHERE NOT EXISTS (
    ->     SELECT 1
    ->     FROM Borrow_Records br
    ->     WHERE br.book_id = b.book_id
    -> );
+---------+-----------------------+-----------+----------------+------------------+-----------+
| book_id | title                 | author_id | published_year | available_copies | genre     |
+---------+-----------------------+-----------+----------------+------------------+-----------+
|       3 | The Alchemist         |         3 |           1988 |                6 | Fiction   |
|       5 | The Discovery of India |         2 |           1946 |                4 | History   |
|       6 | Database Systems      |         1 |           2020 |                5 | Education |
+---------+-----------------------+-----------+----------------+------------------+-----------+
3 rows in set (0.00 sec)

mysql>
```

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